
Literature Review

Discriminative Code Analysis Using Pre-trained Language Models for Framework-Specific Automated Code Review

Research Proposal | March 2026 | Next.js Code Review Specialisation Project

Abstract

Automated code review (ACR) has advanced considerably through the application of pre-trained language models, yet existing systems remain predominantly generative, cloud-dependent, and trained on generic multi-language corpora that inadequately represent the semantics of modern full-stack JavaScript frameworks. This literature review synthesises research across four interconnected areas — automated code review systems, pre-trained code language models, code classification and defect detection, and multi-task learning — to establish the theoretical and empirical foundations for a compact, discriminative code review model specifically designed for Next.js. The review identifies a clear gap at the intersection of framework-specialised review, discriminative multi-task architectures, and local deployment constraints, and articulates four concrete contributions this work makes to address that gap.

Keyword *Automated code review | Discriminative classification | CodeBERT | GraphCodeBERT |*
s: *Multi-task learning | Next.js | Line-level localisation | LoRA fine-tuning*

1. Introduction: The Code Review Problem

Code review — the systematic inspection of source code by one or more developers prior to integration into a production codebase — is among the most widely adopted quality assurance practices in modern software engineering. Empirical evidence consistently demonstrates its value: reviewed code exhibits lower defect rates, higher internal quality, and is generally easier to comprehend and maintain ([Bacchelli & Bird, 2013](#); [McIntosh et al., 2014](#); [Morales et al., 2015](#)). [Bavota & Russo, 2015](#) found that unreviewed code introduces bugs at more than twice the rate of reviewed code, a finding that has been corroborated across multiple industrial and open-source contexts.

Despite its benefits, code review imposes a substantial and often underestimated cost. With more than 28 million software developers globally spending between 10 and 20 percent of their working time on review tasks, the aggregate demand exceeds 22 million developer hours every day. The landmark study by [Bacchelli & Bird \(ICSE 2013\)](#), which surveyed 873 programmers and 165 managers across Microsoft, found that while defect detection is the primary stated motivation for code review, its actual outcomes are considerably broader — encompassing knowledge transfer, team awareness, and the proposal of alternative design solutions. This gap between expected and actual outcomes suggests that human

reviewers are simultaneously overburdened with mechanical checks and underutilised for higher-order reasoning tasks.

Modern large-scale development contexts magnify this tension further. [Sadowski et al. \(ICSE SEIP, 2018\)](#) conducted an extensive study of code review practices at Google and found that developers routinely review hundreds of lines of code per session, yet the quality and depth of review degrades significantly with increasing diff size. Automated tooling that can triage incoming changes, flag likely issues, and localise problematic lines would directly reduce cognitive load and enable reviewers to allocate attention more effectively.

The challenge is further compounded by the specificity of modern frameworks. Next.js, currently the fourth most popular web framework according to the Stack Overflow Developer Survey 2023, introduced a fundamentally new execution model with the App Router in version 13 (stable in version 14, 2023). As noted in the official [Next.js security documentation \(2023\)](#), React Server Components represent a novel paradigm that blurs the line between server and client code execution, creating new categories of defect — including accidental data exposure, incorrect cache directives, misplaced 'use client' boundaries, and insecure Server Action implementations — that have no direct analogue in prior frameworks. No existing automated code review system is trained to recognise or classify these patterns. This is the primary motivation and gap this research addresses.

Research gap

Existing ACR systems are generative, cloud-dependent, or trained on generic multi-language corpora. No compact, discriminative, locally deployable model exists that is specialised for the semantic patterns of the Next.js App Router. This work fills that gap.

2. Automated Code Review: History and State of the Art

Research into automated support for code review can be traced across three broadly distinct generations, distinguished by their underlying methodology and the scope of review tasks they address.

2.1 First generation: Static analysis and rule-based systems

The earliest automated code review tools operated through static analysis — exhaustive enumeration of syntactic and stylistic rules against which source code is checked without execution. Tools such as Checkstyle, FindBugs, and ESLint represent this approach, as does the FindBugs-based ReviewBot system ([Balachandran, 2013](#)) which integrated static analysis into the Gerrit code review platform. As noted by [Singh et al. \(2017\)](#), static analysis tools provide precise, reproducible feedback on well-defined rule violations but are fundamentally

incapable of reasoning about context, semantics, or intent. A rule engine cannot determine that a particular use of `fetch()` without a cache directive is problematic specifically because the enclosing component is a Next.js Server Component — such reasoning requires understanding of both the code and its execution environment.

Additionally, most code review platforms — including Gerrit, Phabricator, and Code Flow — provide only basic automation support and do not reduce reviewers' burden in a meaningful way (Tymchuk *et al.*, 2015; Pascarella *et al.*, 2018). This gap between the available tooling and the needs of modern development teams motivated the transition to learning-based approaches.

2.2 Second generation: Neural machine translation models

The second generation of ACR systems reframed code review as a machine translation problem — specifically, treating the transformation of a code diff into either a reviewer comment or a corrected code version as a sequence-to-sequence learning task. Tufano *et al.* (ICSE 2021) were among the first to frame both the contributor and reviewer perspectives of code review as neural machine translation tasks, using Transformer architectures trained on real GitHub pull request data. Their work established the foundational insight that reviewer comments and code diffs constitute naturally paired training data, making large-scale supervised learning feasible without manual annotation.

DeepReview (Lo, PAKDD 2019) introduced multi-instance learning for review classification, providing an early demonstration that discriminative models — models that predict a class rather than generate text — could be trained on review corpora with meaningful accuracy. This work is a direct intellectual predecessor of the discriminative architecture proposed in the present research.

2.3 Third generation: Pre-trained transformer models

The defining work of the current generation is **CodeReviewer** (Li *et al.*, ESEC/FSE 2022), which introduced large-scale pre-training on code review data.

“CodeReviewer defines three canonical ACR sub-tasks — code change quality estimation (does this diff need a review?), review comment generation (what should the reviewer say?), and code refinement (given a comment, fix the code) — and pre-trains a transformer encoder-decoder on a large multilingual dataset of GitHub pull requests spanning nine programming languages, establishing benchmarks that have become the standard reference point for subsequent work.”

Critically for the present work, the **code change quality estimation** sub-task is a discriminative binary classification problem — precisely the architecture proposed here — and CodeReviewer demonstrates that this task is tractable from PR data alone, without requiring manual labelling beyond the signal implicit in whether a reviewer left a comment.

Subsequent work has explored a range of extensions. Tufano *et al.* (ASE 2022) demonstrated that pre-trained models substantially improve over raw Transformer

baselines on all three CodeReviewer tasks. [Lu et al. \(LLaMA-Reviewer, 2023\)](#) applied LoRA fine-tuning to LLaMA-series models, supporting all three ACR tasks without full retraining. [Carllm \(ACM Transactions, 2024\)](#) fine-tuned LLaMA with chain-of-thought prompting on 1,793 open-source projects, demonstrating that comprehensibility — not just accuracy — improves with fine-tuning, a finding with direct implications for the design of user-facing outputs in this work. [Haider et al. \(2024\)](#) applied QLoRA to CodeLlama and Llama 3.1 for review comment generation, reporting 25–83% improvement over untuned baselines, further validating the PEFT approach for code review tasks.

A significant practical limitation of all third-generation systems is their cloud dependency. [Cihan et al. \(2024\)](#), in an industrial study of an LLM-based review tool deployed across 238 practitioners and 4,335 pull requests, found that while such tools complement human review in principle, cloud infrastructure requirements introduce latency, data privacy risks, and cost barriers that limit adoption — particularly in regulated industries and for proprietary codebases. This observation directly motivates the offline, local deployment model proposed in the present work.

2.4 Limitations of current approaches for Next.js

All extant ACR systems share a common limitation with respect to the target domain of this research: they are trained on generic, multi-language corpora in which Next.js-specific patterns are statistically negligible. The App Router paradigm, introduced in Next.js 13 (2022) and stabilised in version 14 (2023), introduced concepts — Server Components, Client Components, Server Actions, the 'use client' boundary, revalidate cache semantics, and route segment configuration — that require specialised semantic understanding. A model that has never encountered these patterns in training cannot be expected to classify their misuse reliably, regardless of its general code understanding capability. This gap is compounded by the rapid pace of Next.js versioning: App Router patterns that were best practice in version 13 differ meaningfully from those in version 15, and general-purpose models trained before 2023 have no exposure to these patterns at all.

3. Pre-trained Code Language Models

The proposed discriminative architecture requires a strong, compact code representation backbone. This section surveys the relevant pre-trained models, their architectural properties, and the empirical evidence bearing on their suitability as a fine-tuning base for the classification tasks proposed here.

3.1 CodeBERT: bimodal pre-training for code and language

The foundational model for this work is **CodeBERT** ([Feng et al., Findings of EMNLP 2020, arXiv:2002.08155](#)), a bimodal pre-trained model based on the RoBERTa architecture, trained on a corpus of 2.1 million bimodal NL-PL pairs (documentation

paired with code) and 6.4 million unimodal code samples across six programming languages, using a hybrid objective combining masked language modelling and replaced token detection.

“CodeBERT learns general-purpose representations that support downstream NL-PL applications. We train CodeBERT with a hybrid objective function that incorporates replaced token detection, which enables the model to utilise both bimodal data of NL-PL pairs and unimodal data.”

With 125 million parameters and a serialised size of approximately 500 MB — reducible to under 200 MB with 4-bit quantisation — CodeBERT is the most practically deployable of the candidate encoders. Fine-tuned for binary classification on the CodeXGLUE defect detection benchmark (Devign dataset), a CodeBERT-based model achieves 97.2% accuracy, 98.05% precision, 97.31% recall, and an F1-score of 97.68%, outperforming earlier deep learning baselines by a substantial margin (*Code Vulnerability Detection study, arXiv:2508.11710*).

3.2 GraphCodeBERT: structural semantics via data flow

GraphCodeBERT (Guo et al., ICLR 2021, *arXiv:2009.08366*) extends CodeBERT by incorporating semantic-level structural information — specifically, data flow graphs derived from abstract syntax trees — into the pre-training objective. This allows the model to capture variable dependency chains and data flow relationships that are invisible to purely token-based encoders.

“GraphCodeBERT overcomes the limitation of treating code fragments solely as token sequences. By leveraging data flow — which encodes how values flow between variables — the model can capture semantic-level structural properties of code that are relevant for tasks requiring deeper program understanding.”

For the task of detecting performance and logic issues in Next.js code, where understanding the flow of data between Server Components, Client Components, and API route handlers is essential, GraphCodeBERT's structural awareness represents a meaningful advantage over CodeBERT. The two models constitute the primary ablation comparison in the proposed experimental design. Research applying supervised contrastive learning (SCL-CVD) to GraphCodeBERT with LoRA fine-tuning on three real-world vulnerability datasets (Wang et al., *Computers and Security* 2024) reports a relative improvement of 35.68% over baseline approaches, while LoRA introduces neither inference latency nor input sequence length constraints — a critical property for local deployment.

3.3 Layer-wise representation properties and fine-tuning behaviour

A key question for the proposed training approach concerns which model layers to fine-tune and which to freeze. Shi et al. (*arXiv:2304.05216, 2023*) conducted a systematic empirical study of CodeBERT's internal representations, finding that lexical, syntactic, and structural properties of source code are progressively encoded in lower, intermediate, and higher layers respectively, while semantic

properties are distributed across the entire model. Crucially, fine-tuning preserves most pre-trained code properties — the model does not catastrophically forget its pre-trained code knowledge when adapted to a downstream classification task.

This finding has two practical implications for this work. First, it supports a strategy of freezing the lower encoder layers (which encode general token-level code structure) and fine-tuning only the upper layers and task-specific heads, reducing training cost without sacrificing code understanding quality. Second, it supports the use of LoRA adapters (Hu et al., 2021, arXiv:2106.09685) applied specifically to the attention weight matrices in the upper layers, where the bulk of task-specific semantic adaptation occurs. A systematic literature review on parameter-efficient fine-tuning for code models confirms that defect detection using LoRA on CodeBERT and GraphCodeBERT achieves strong results on established benchmarks with substantially reduced parameter counts (*PEFT survey for code models, 2024*).

3.4 Alternative backbone candidates

Model	Params	Architecture	Key property	Suitability
CodeBERT	125 M	RoBERTa encoder	Bimodal NL+PL pre-training	Primary candidate
GraphCodeBERT	125 M	RoBERTa + data flow	Structural code semantics	Ablation baseline
CodeBERTa-small	84 M	RoBERTa (smaller)	Faster inference	Speed comparison
CodeT5 (enc. only)	60 M	T5 encoder	Identifier-aware	Secondary candidate
UniXcoder	125 M	Unified cross-modal	Code + AST + NL unified	Future evaluation

Table 1. Candidate encoder models for the discriminative backbone. All models are compatible with the [CLS] token classification paradigm and the 512-token input length available within the standard BERT context window.

4. Code Classification and Defect Detection

The multi-label issue category head at the core of the proposed architecture is an instance of the broader family of code classification problems, for which a substantial and mature body of empirical evidence exists. This section surveys the most directly relevant results, from function-level defect detection to fine-grained line-level prediction.

4.1 Function-level defect and vulnerability detection

The CodeXGLUE benchmark (Lu et al., *NeurIPS Datasets and Benchmarks, 2021*) established a standard evaluation suite for code intelligence tasks, including defect

detection. Research combining graph neural networks with pre-trained transformer encoders for software defect detection on the CodeXGLUE Defect sub-task achieves F1-scores of up to 91.7%, representing a 23.6-point absolute improvement over LSTM-based baselines — demonstrating that pre-trained transformer encoders substantially outperform earlier sequence models on code classification.

For the more specific task of vulnerability detection, the contrastive learning approach **SCL-CVD** (Wang et al., *Computers & Security*, 2024) fine-tunes GraphCodeBERT using LoRA on three widely used benchmark datasets — Devign, Reveal, and Big-Vul — and reports a relative improvement of 35.68% over all baselines, including vanilla CodeBERT and GraphCodeBERT fine-tuned with standard cross-entropy loss. The paper attributes this improvement to the contrastive objective's ability to separate the representations of vulnerable and non-vulnerable samples in the embedding space, which is particularly effective when the two classes are superficially similar — a characteristic shared by the issue categories targeted in this work, where a correctly structured Server Component and one with a subtle cache misconfiguration are lexically nearly identical.

4.2 Multi-label classification for code quality

The issue category head proposed here produces multi-label outputs: a single diff may receive multiple category labels simultaneously. This differs from the binary classification setting of most defect detection research. The technical approach is well-established: replacing the binary sigmoid output of a standard classifier with a multi-label sigmoid head trained with binary cross-entropy loss per class, with a shared encoder whose [CLS] token representation serves as the aggregate code representation. This formulation is identical to that used in multi-label text classification tasks in NLP, and has been demonstrated to transfer effectively to code classification settings with code-aware encoders (CodeXGLUE, Lu et al., 2021).

A key practical challenge in multi-label code classification is class imbalance: issues such as security vulnerabilities are rare compared to naming or performance issues. Research on vulnerability detection consistently identifies imbalance as a critical factor (Cheshkov et al., *ICCQ* 2024), and prior work recommends focal loss or sample weighting strategies to address this. The label extraction methodology proposed in this work (extracting labels from existing reviewer comments using a pre-classification step) must account for this imbalance during both collection and training.

4.3 Line-level prediction: the LineVul precedent

The line-level localisation head proposed in this architecture draws directly from **LineVul** (Fu & Tantithamthavorn, *MSR* 2022, DOI:10.1145/3524842.3528452), the first transformer-based system to predict software vulnerabilities at the line level rather than the function or file level.

“LineVul achieves an F-measure of 0.91 — 160% to 379% better than state-of-the-art approaches, with a median improvement of 250%. It achieves Precision of 0.97 and Recall of 0.86, outperforming baseline approaches by 322% and 19% respectively. Top-10 Accuracy for line-level predictions is 0.65, 12%–25% more accurate than prior approaches, evaluated on a large-scale real-world dataset of 188,000+ C/C++ functions.”

LineVul's architecture — a CodeBERT encoder whose token-level output representations are passed to a token classification head with attention-based line scoring — is the direct architectural precedent for the line-level head proposed here. The key adaptation for this work is the domain shift from C/C++ vulnerability prediction to TypeScript/Next.js code review: rather than predicting whether a line contains a security vulnerability, the head predicts whether a line represents a code review issue of any category. The training signal for this head is derived from the line numbers attached to GitHub inline review comments, which are available in the raw PR data at no additional annotation cost.

LineVul's empirical finding that transformer-based line-level prediction is accurate across the Top-25 most dangerous CWE categories (75%–100% true positive rate) provides strong prior evidence that the underlying approach generalises beyond the specific training distribution — a property that will be tested in this work across Next.js-specific issue categories.

5. Multi-Task Learning for Shared Encoder Architectures

The proposed model trains a single shared CodeBERT/GraphCodeBERT encoder simultaneously across four tasks: multi-label issue category classification, quality score regression, severity classification, and line-level localisation. This multi-task learning (MTL) design is grounded in a substantial body of theoretical and empirical work in both general NLP and code-specific contexts.

5.1 Foundations: MT-DNN and the regularisation effect of MTL

The canonical reference for shared-encoder MTL in NLP is **MT-DNN** ([Liu et al., ACL 2019, arXiv:1901.11504](#)), a Microsoft Research contribution that applies multi-task learning across all nine tasks of the GLUE benchmark using a shared BERT encoder with task-specific output layers.

“The lower layers are shared across all tasks while the top layers are task-specific. The input is first represented as a sequence of embedding vectors in I_1 . Then the Transformer encoder captures the contextual information for each word and generates shared contextual embedding vectors in I_2 . Finally, for each task, additional task-specific layers generate task-specific representations, followed by operations necessary for classification, similarity scoring, or relevance ranking.”

MT-DNN produces state-of-the-art results on 8 of the 9 GLUE tasks, and subsequently achieves new state-of-the-art results on SNLI (91.6%, a 1.5-point

absolute improvement) and SciTail (95.0%, a 6.7-point absolute improvement). Crucially, the performance gains are largest on tasks with smaller datasets — directly relevant for this project, where training data for rare issue categories (e.g., [security](#)) will be scarcer than for common ones (e.g., [naming](#)).

The theoretical basis for this behaviour is the [regularisation effect](#) of multi-task learning: because the shared encoder must simultaneously represent useful features for multiple tasks, it is prevented from overfitting to idiosyncrasies of any single task's training distribution. As [Caruana \(1997\)](#) originally characterised it, multi-task learning uses the domain information contained in related tasks' training signals as an inductive bias, improving generalisation by learning tasks in parallel while sharing representation. For the present work, issue category, severity, and quality score are strongly correlated signals: a security issue is almost always critical, while a naming issue is almost invariably informational. Learning these joint distributions produces shared representations that are more discriminative for each individual task than single-task fine-tuning would achieve.

5.2 Negative transfer and task weighting

Multi-task learning does not unconditionally improve performance. When the gradients of different tasks conflict — pointing in opposing directions or differing significantly in magnitude — joint training can hinder individual task performance, a phenomenon known as [negative transfer](#) ([Wikipedia, Multi-task learning](#); [Chen et al., Semantic Scholar 2021](#)). For the proposed architecture, the risk of negative transfer is low because the four tasks are genuinely correlated: all operate on the same input (code diff + context), and their outputs (category, severity, quality, line location) are jointly determined by the same underlying properties of the code change. Nonetheless, the training objective must weight task losses appropriately:

The combined training loss is defined as $L = w_1 * L_{cat} + w_2 * L_{quality} + w_3 * L_{severity} + w_4 * L_{lines}$, where w_1, w_2, w_3, w_4 are task weights and $L_{cat}, L_{quality}, L_{severity}$, and L_{lines} are the per-task losses. Initial experiments will use equal weights (all $w = 1.0$), followed by grid search to identify the optimal balance. Uncertainty weighting ([Kendall et al., CVPR 2018](#)), which adaptively weights task losses by learned homoscedastic uncertainty, will be explored as an alternative to manual tuning.

5.3 MTL applied to code understanding tasks

The application of multi-task learning specifically to code intelligence is supported by a growing body of empirical work. Research combining dual-channel architectures with shared semantic encoders for defect detection shows that joint training on related sub-tasks produces representations that generalise better than single-task fine-tuning. [Multi-task learning in biomedical NLP \(Springer 2023\)](#) demonstrates that MT-DNN-style architectures with task-specific heads achieve superior results compared to single-task models even when the per-task dataset is

small, which directly supports the proposed approach for low-frequency issue categories.

6. Input Representation and Contextual Enrichment

The proposed input strategy concatenates the code diff with surrounding file context using the BERT-standard segment structure: [CLS] <diff> [SEP] <context> [SEP]. This section reviews the evidence for this design choice and its specific necessity in the Next.js domain.

6.1 Empirical evidence for context-enriched representations

A study presented at MSR 2024 found that encoding structural context and version history alongside source code improves code classification model performance across five pre-trained architectures — including CodeBERT and GraphCodeBERT — with contextual signals providing consistent gains beyond what static code snapshots alone can achieve. While that work focused primarily on file-level structural context and commit history, the principle generalises: models that observe more of the execution environment in which code operates make better classification decisions.

The technical basis for this approach is the BERT sentence-pair input format, in which the [CLS] token aggregates information from both segments through bidirectional self-attention. This enables the encoder to learn joint representations of the diff and its surrounding context simultaneously, rather than processing them independently and fusing them post-hoc. The 512-token input limit of CodeBERT/GraphCodeBERT constrains the amount of context that can be incorporated; the proposed approach uses 30 lines around changed lines, which empirically fits within the token budget for the majority of TypeScript function bodies.

6.2 Framework-specific context: why it is non-optional

For the Next.js domain specifically, file context is not merely beneficial — it is necessary for correct classification of a substantial fraction of issues. Three concrete examples illustrate this:

Issue category	Information missing from diff alone	Available in file context
nextjs-pattern	Whether a component is a Server or Client Component	'use client' directive at file top
security	Whether a Server Action validates its caller	'use server' annotation and auth checks
performance	Whether fetch() is inside a cached route segment	export const revalidate in layout.ts

Table 2. Issue categories that require file context for correct classification. Without surrounding context, all three diffs are ambiguous or appear benign.

Additionally, the file path itself — for example, `app/dashboard/page.tsx` versus `components/ui/button.tsx` — is a lightweight but highly informative contextual signal that disambiguates App Router page components from general UI components. As described in the official [Next.js production checklist documentation](#), correct placement of 'use client' boundaries, appropriate use of Dynamic APIs, and correct Server Action implementation are all context-dependent. The file path is prepended as a prefix token before the diff, costing fewer than 10 tokens and providing material disambiguation.

7. Research Gap and Contribution

7.1 Synthesis of the identified gap

The literature reviewed above establishes five independently supported propositions that together define the research gap this work addresses:

P 1	Code review is expensive and reviewer attention is the binding constraint on throughput (Bacchelli & Bird, 2013; Sadowski et al., 2018).
P 2	Discriminative classification models trained on review data can accurately predict code quality issues (CodeReviewer quality estimation sub-task; DeepReview, 2019).
P 3	Compact pre-trained code encoders (CodeBERT, GraphCodeBERT) are highly effective for code classification tasks and deploy feasibly on commodity hardware (Feng et al., 2020; Guo et al., 2021; LineVul, 2022).
P 4	Multi-task learning across correlated tasks improves individual task performance, particularly in limited-data regimes (MT-DNN, Liu et al., 2019).
P 5	Next.js App Router introduces semantically distinct patterns that are absent from all existing ACR training corpora and require framework-specialised reasoning (Next.js documentation, 2023).

From P1–P5, the gap follows directly: an architecture satisfying P2–P4 and trained on data capturing P5 does not yet exist. Existing systems satisfy at most three of the five propositions simultaneously: CodeReviewer satisfies P1 and P2 but not P3, P4, or P5; commercial systems (Qodo, Coderabbit, GitHub Copilot Code Review) satisfy P1 and P2 but are generative, cloud-dependent, and framework-generic. No existing system satisfies all five simultaneously.

7.2 Contributions of this work

Against this landscape, this research makes the following four contributions:

C 1	A discriminative multi-task architecture	A CodeBERT/GraphCodeBERT encoder fine-tuned with four task-specific heads (multi-label category, quality regression, severity classification, line-level localisation) that runs entirely locally on commodity hardware without GPU acceleration, achieving sub-200ms inference per diff.
C 2	A Next.js-specialised annotated dataset	A curated dataset of pull request diffs from high-activity Next.js repositories (vercel/next.js, calcom/cal.com, shadcn-ui/ui, and others) with extracted multi-label annotations covering eight issue categories derived from inline reviewer comments.
C 3	A systematic encoder comparison	An ablation study comparing CodeBERT and GraphCodeBERT as encoder backbones across all four task heads and two input strategies (diff-only vs. diff + file context), establishing which architectural choices drive performance on Next.js-specific issue categories.
C 4	Line-level output localisation	A token classification head that predicts which specific lines within a diff are problematic, providing developer-actionable output at a granularity that current compact code review tools do not support — trained using line numbers from GitHub inline comments as supervision.

8. Conclusion

This literature review has surveyed four areas of directly relevant prior work — automated code review systems, pre-trained code language models, code classification and defect detection, and multi-task learning — and established the theoretical and empirical foundations for the proposed research. The review demonstrates that all component techniques are individually well-validated: discriminative code classification with CodeBERT achieves F1 scores above 0.91 on established benchmarks; multi-task learning with shared encoders consistently improves over single-task baselines, especially in limited-data regimes; line-level prediction is tractable with transformer-based token classification heads; and context-enriched input representations improve code classification accuracy across a range of pre-trained architectures.

The gap identified by this review is not merely an absence of prior work but a principled confluence: the existing literature has not combined these techniques in a locally deployable, framework-specialised discriminative architecture. Next.js in particular represents an underexplored but high-value target domain whose specific patterns — Server Components, App Router semantics, caching directives, and Server Actions — are semantically distinct from prior-art training corpora and require dedicated treatment. The four contributions proposed in this research directly address this gap, and the technical feasibility of each contribution is well-supported by the reviewed literature.

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